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Proper and improper constructions in Pavel Tichý's transparent intensional logic.

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This presentation analyzes the theoretical distinction between proper and improper constructions within the framework of Transparent inequality Logic (TIL), a logical system developed by the Czech logician and philosopher Pavel Tichý. Unlike Fregean logic, which mediates reference through Sinn, in TIL constructions (variable, trivialization, execution, double execution, composition, and closure) are objectively structured procedures (they are not mental and do not depend on language) that produce entities from a ramified type base. From this point of view, we move from a two-level semantics (intension-extension) to a three-level one: (construction-intension-extension). The problem we address is the distinction of improper constructions and proper constructions. In this sense, we argue that both are subject to the same procedures, although the result of one and the other is different. For example, in $x + 3 = 5$, the variable (construction) x constructs 2 (it is a proper construction), while it constructs nothing in $x + 0 = 1$ (improper construction); (what about the construction 'the perfect odd number'?). We also examine Tichý's critique of the distinction between use and mention, proposing that his notion of construction offers an adequate framework for explaining substitutivity in contexts of *de dicto* and *de re*. We conclude by showing that the distinction between proper and improper construction is fundamental to the treatment of the *what* and *how* of logical-conceptual analysis.

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